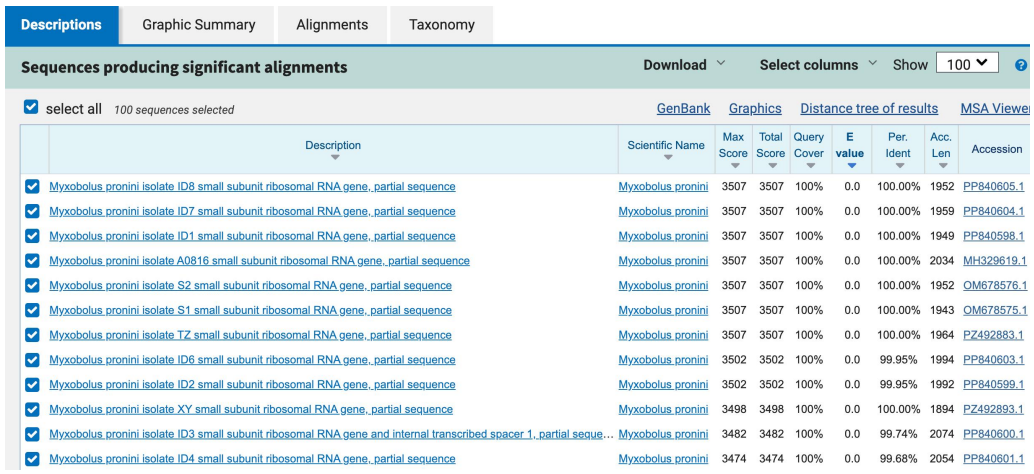
A. BLAST-based identification of four isolates



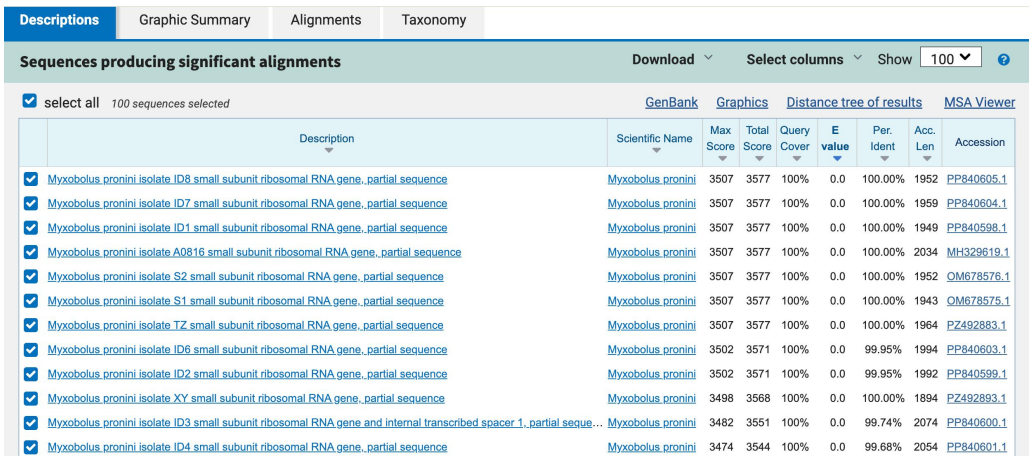
Descriptions | Graphic Summary | Alignments | Taxonomy

Sequences producing significant alignments — Download | Select columns | Show 100

select all *100 sequences selected* — GenBank | Graphics | Distance tree of results | MSA Viewer

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
Myxobolus pronini isolate ID8 small subunit ribosomal RNA gene, partial sequence	Myxobolus pronini	3507	3507	100%	0.0	100.00%	1952	PP840605.1
Myxobolus pronini isolate ID7 small subunit ribosomal RNA gene, partial sequence	Myxobolus pronini	3507	3507	100%	0.0	100.00%	1959	PP840604.1
Myxobolus pronini isolate ID1 small subunit ribosomal RNA gene, partial sequence	Myxobolus pronini	3507	3507	100%	0.0	100.00%	1949	PP840598.1
Myxobolus pronini isolate A0816 small subunit ribosomal RNA gene, partial sequence	Myxobolus pronini	3507	3507	100%	0.0	100.00%	2034	MH329619.1
Myxobolus pronini isolate S2 small subunit ribosomal RNA gene, partial sequence	Myxobolus pronini	3507	3507	100%	0.0	100.00%	1952	OM678576.1
Myxobolus pronini isolate S1 small subunit ribosomal RNA gene, partial sequence	Myxobolus pronini	3507	3507	100%	0.0	100.00%	1943	OM678575.1
Myxobolus pronini isolate TZ small subunit ribosomal RNA gene, partial sequence	Myxobolus pronini	3507	3507	100%	0.0	100.00%	1964	PZ492883.1
Myxobolus pronini isolate ID6 small subunit ribosomal RNA gene, partial sequence	Myxobolus pronini	3502	3502	100%	0.0	99.95%	1994	PP840603.1
Myxobolus pronini isolate ID2 small subunit ribosomal RNA gene, partial sequence	Myxobolus pronini	3502	3502	100%	0.0	99.95%	1992	PP840599.1
Myxobolus pronini isolate XY small subunit ribosomal RNA gene, partial sequence	Myxobolus pronini	3498	3498	100%	0.0	100.00%	1894	PZ492893.1
Myxobolus pronini isolate ID3 small subunit ribosomal RNA gene and internal transcribed spacer 1, partial seque…	Myxobolus pronini	3482	3482	100%	0.0	99.74%	2074	PP840600.1
Myxobolus pronini isolate ID4 small subunit ribosomal RNA gene, partial sequence	Myxobolus pronini	3474	3474	100%	0.0	99.68%	2054	PP840601.1

Myxobolus pronini XY



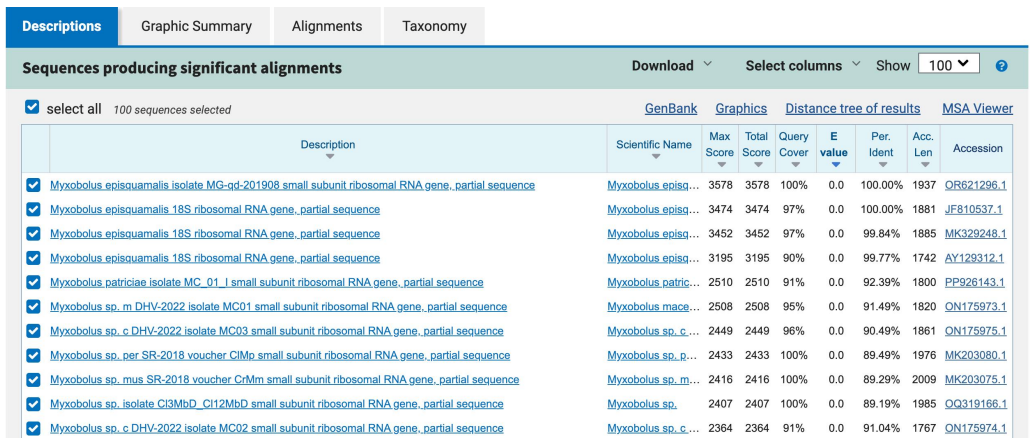
Descriptions | Graphic Summary | Alignments | Taxonomy

Sequences producing significant alignments — Download | Select columns | Show 100

select all *100 sequences selected* — GenBank | Graphics | Distance tree of results | MSA Viewer

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
Myxobolus pronini isolate ID8 small subunit ribosomal RNA gene, partial sequence	Myxobolus pronini	3507	3577	100%	0.0	100.00%	1952	PP840605.1
Myxobolus pronini isolate ID7 small subunit ribosomal RNA gene, partial sequence	Myxobolus pronini	3507	3577	100%	0.0	100.00%	1959	PP840604.1
Myxobolus pronini isolate ID1 small subunit ribosomal RNA gene, partial sequence	Myxobolus pronini	3507	3577	100%	0.0	100.00%	1949	PP840598.1
Myxobolus pronini isolate A0816 small subunit ribosomal RNA gene, partial sequence	Myxobolus pronini	3507	3577	100%	0.0	100.00%	2034	MH329619.1
Myxobolus pronini isolate S2 small subunit ribosomal RNA gene, partial sequence	Myxobolus pronini	3507	3577	100%	0.0	100.00%	1952	OM678576.1
Myxobolus pronini isolate S1 small subunit ribosomal RNA gene, partial sequence	Myxobolus pronini	3507	3577	100%	0.0	100.00%	1943	OM678575.1
Myxobolus pronini isolate TZ small subunit ribosomal RNA gene, partial sequence	Myxobolus pronini	3507	3577	100%	0.0	100.00%	1964	PZ492883.1
Myxobolus pronini isolate ID6 small subunit ribosomal RNA gene, partial sequence	Myxobolus pronini	3502	3571	100%	0.0	99.95%	1994	PP840603.1
Myxobolus pronini isolate ID2 small subunit ribosomal RNA gene, partial sequence	Myxobolus pronini	3502	3571	100%	0.0	99.95%	1992	PP840599.1
Myxobolus pronini isolate XY small subunit ribosomal RNA gene, partial sequence	Myxobolus pronini	3498	3568	100%	0.0	100.00%	1894	PZ492893.1
Myxobolus pronini isolate ID3 small subunit ribosomal RNA gene and internal transcribed spacer 1, partial seque…	Myxobolus pronini	3482	3551	100%	0.0	99.74%	2074	PP840600.1
Myxobolus pronini isolate ID4 small subunit ribosomal RNA gene, partial sequence	Myxobolus pronini	3474	3544	100%	0.0	99.68%	2054	PP840601.1

Myxobolus pronini TZ



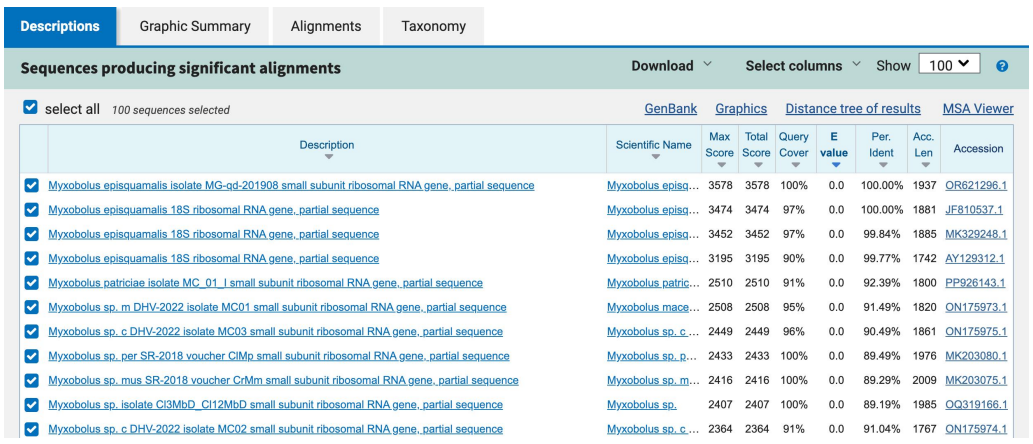
Descriptions | Graphic Summary | Alignments | Taxonomy

Sequences producing significant alignments — Download | Select columns | Show 100

select all *100 sequences selected* — GenBank | Graphics | Distance tree of results | MSA Viewer

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
Myxobolus episquamalis isolate MG-qd-201908 small subunit ribosomal RNA gene, partial sequence	Myxobolus episq…	3578	3578	100%	0.0	100.00%	1937	OR621296.1
Myxobolus episquamalis 18S ribosomal RNA gene, partial sequence	Myxobolus episq…	3474	3474	97%	0.0	100.00%	1881	JF810537.1
Myxobolus episquamalis 18S ribosomal RNA gene, partial sequence	Myxobolus episq…	3452	3452	97%	0.0	99.84%	1885	MK329248.1
Myxobolus episquamalis 18S ribosomal RNA gene, partial sequence	Myxobolus episq…	3195	3195	90%	0.0	99.77%	1742	AY129312.1
Myxobolus patriciae isolate MC_01_I small subunit ribosomal RNA gene, partial sequence	Myxobolus patric…	2510	2510	91%	0.0	92.39%	1800	PP926143.1
Myxobolus sp. m DHV-2022 isolate MC01 small subunit ribosomal RNA gene, partial sequence	Myxobolus mace…	2508	2508	95%	0.0	91.49%	1820	ON175973.1
Myxobolus sp. c DHV-2022 isolate MC03 small subunit ribosomal RNA gene, partial sequence	Myxobolus sp. c…	2449	2449	96%	0.0	90.49%	1861	ON175975.1
Myxobolus sp. per SR-2018 voucher ClMp small subunit ribosomal RNA gene, partial sequence	Myxobolus sp. p…	2433	2433	100%	0.0	89.49%	1976	MK203080.1
Myxobolus sp. mus SR-2018 voucher CrMm small subunit ribosomal RNA gene, partial sequence	Myxobolus sp. m…	2416	2416	100%	0.0	89.29%	2009	MK203075.1
Myxobolus sp. isolate Cl3MbD_Cl12MbD small subunit ribosomal RNA gene, partial sequence	Myxobolus sp.	2407	2407	100%	0.0	89.19%	1985	OQ319166.1
Myxobolus sp. c DHV-2022 isolate MC02 small subunit ribosomal RNA gene, partial sequence	Myxobolus sp. c…	2364	2364	91%	0.0	91.04%	1767	ON175974.1

Myxobolus episquamalis QD



Descriptions | Graphic Summary | Alignments | Taxonomy

Sequences producing significant alignments — Download | Select columns | Show 100

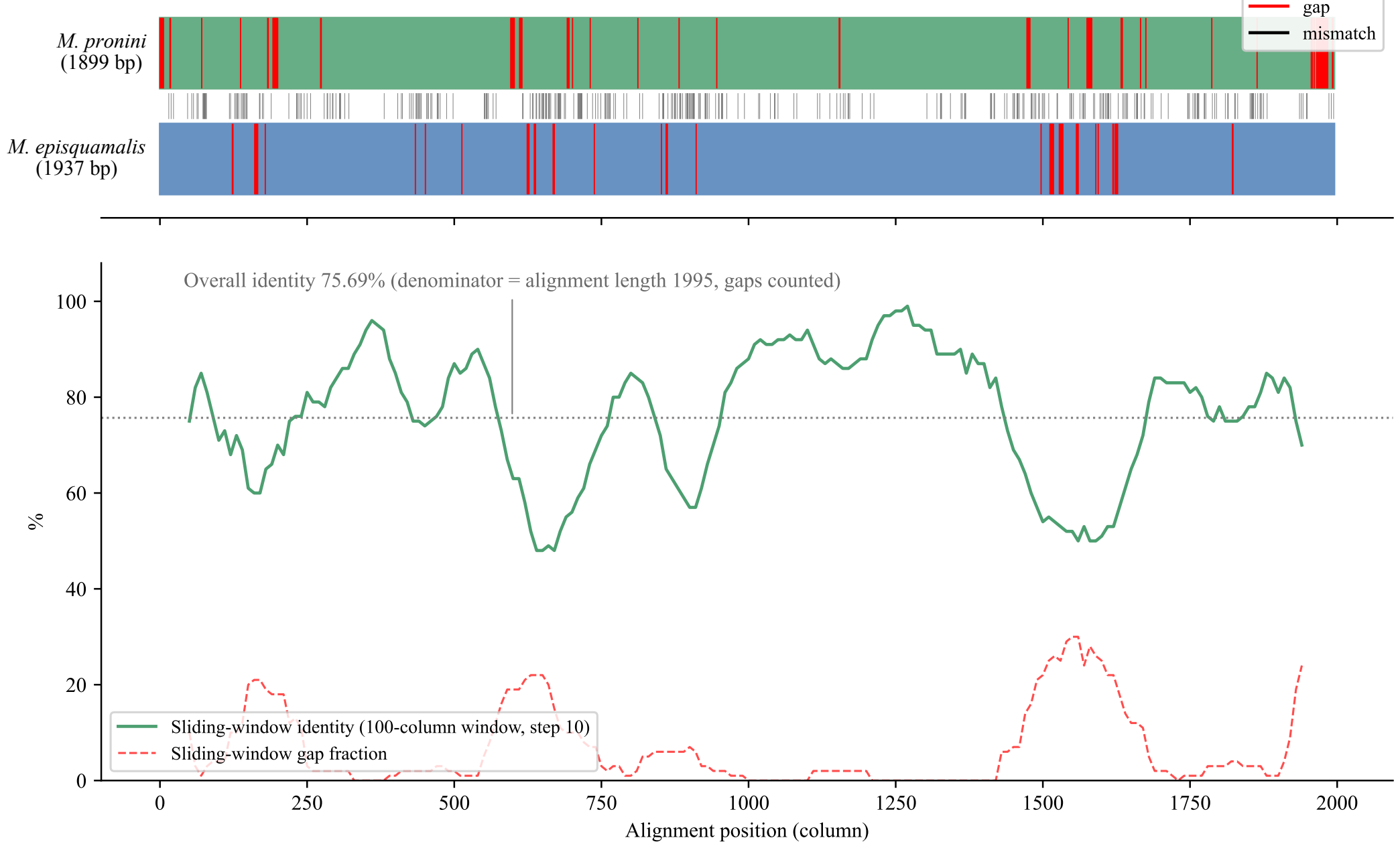
select all *100 sequences selected* — GenBank | Graphics | Distance tree of results | MSA Viewer

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
Myxobolus episquamalis isolate MG-qd-201908 small subunit ribosomal RNA gene, partial sequence	Myxobolus episq…	3578	3578	100%	0.0	100.00%	1937	OR621296.1
Myxobolus episquamalis 18S ribosomal RNA gene, partial sequence	Myxobolus episq…	3474	3474	97%	0.0	100.00%	1881	JF810537.1
Myxobolus episquamalis 18S ribosomal RNA gene, partial sequence	Myxobolus episq…	3452	3452	97%	0.0	99.84%	1885	MK329248.1
Myxobolus episquamalis 18S ribosomal RNA gene, partial sequence	Myxobolus episq…	3195	3195	90%	0.0	99.77%	1742	AY129312.1
Myxobolus patriciae isolate MC_01_I small subunit ribosomal RNA gene, partial sequence	Myxobolus patric…	2510	2510	91%	0.0	92.39%	1800	PP926143.1
Myxobolus sp. m DHV-2022 isolate MC01 small subunit ribosomal RNA gene, partial sequence	Myxobolus mace…	2508	2508	95%	0.0	91.49%	1820	ON175973.1
Myxobolus sp. c DHV-2022 isolate MC03 small subunit ribosomal RNA gene, partial sequence	Myxobolus sp. c…	2449	2449	96%	0.0	90.49%	1861	ON175975.1
Myxobolus sp. per SR-2018 voucher ClMp small subunit ribosomal RNA gene, partial sequence	Myxobolus sp. p…	2433	2433	100%	0.0	89.49%	1976	MK203080.1
Myxobolus sp. mus SR-2018 voucher CrMm small subunit ribosomal RNA gene, partial sequence	Myxobolus sp. m…	2416	2416	100%	0.0	89.29%	2009	MK203075.1
Myxobolus sp. isolate Cl3MbD_Cl12MbD small subunit ribosomal RNA gene, partial sequence	Myxobolus sp.	2407	2407	100%	0.0	89.19%	1985	OQ319166.1
Myxobolus sp. c DHV-2022 isolate MC02 small subunit ribosomal RNA gene, partial sequence	Myxobolus sp. c…	2364	2364	91%	0.0	91.04%	1767	ON175974.1

Myxobolus episquamalis XM

B. Pairwise 18S rDNA alignment and sequence identity

Needleman-Wunsch global alignment: match +2; mismatch -1; gap opening -5; gap extension -0.5.



Alignment: 1,995 columns **Matches:** 1,510 **Mismatches:** 331 **Gap columns:** 154

Sequence identity: 75.69% including gap columns (1,510/1,995); 82.02% excluding gap columns (1,510/1,841).

Fig. S2. 18S rDNA-based identification and sequence comparison